Katherine (Kat) L. Titterton

CURRICULUM VITAE

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PROFESSIONAL SUMMARY

Interdisciplinary scientist and engineer working at the interface of computational imaging, machine learning, and laboratory automation in biotechnology. Eight-plus years designing and scaling high-content / high-throughput biological assays and the image- and data-analysis systems that turn them into machine-learning training data. Founding member of the technical staff at an AI-driven drug-safety company; inventor of novel phenotypic bioactivity-detection methods; co-author of eight peer-reviewed papers in journals including Nature Communications, Cell Systems, iScience, the Journal of Biological Chemistry, and Molecular Neurodegeneration Advances, including studies establishing new benchmarks for small-molecule bioactivity prediction.

SELECTED CONTRIBUTIONS OF MAJOR SIGNIFICANCE

- Invented **two novel bioactivity (morphology-change) detection methods** that sensitively detect small-molecule effects on cells at approximately **10× lower concentrations** than conventional direct-toxicity assays (LDH, MT-Glo, cell count), expanding the sensitivity range of phenotypic safety screening.
- Oversaw assay development and data generation for **two of the world's largest Cell Painting datasets** (100,000+ unique compounds), foundational training data for in silico toxicology models.
- Co-developed an interpretable **in silico clinical-risk model for drug-induced liver injury (DILI)** reaching **0.92 AUC at 77% sensitivity and 90% specificity**, matching or exceeding industry-standard in vitro assays (presented at the Society of Toxicology Annual Meeting, 2025).
- Co-authored a field-influencing study showing that several widely used **small-molecule bioactivity benchmarks are largely predicted by simply counting cells**, reframing how the field validates phenotypic models (Nature Communications, 2026).
- Optimized multiplexed expansion revealing (mExR) enabling simultaneous nanoscale imaging of **30+ proteins** across 16× expanded brain tissue (published in Nature Communications, 2024).

PROFESSIONAL EXPERIENCE

Founding Member of the Technical Staff, Axiom Bio *2023 – Present*
San Francisco, CA · AI-driven drug-safety / new approach methodologies (NAMs)

- Founding technical hire developing new approach methodologies enabling medicinal chemists to characterize the bioactivity profile of their molecules and design safer drugs.
- Led feature engineering and algorithm development; oversaw assay development and data generation for two of the world's largest Cell Painting datasets (100,000+ compounds).
- Invented two novel bioactivity-detection methods (~10× more sensitive than conventional toxicity assays); contributed to an in silico DILI clinical-risk model (0.92 AUC).
- Drove scientific product direction and pivot decisions.

Associate Scientist III, Insitro *2022 – 2023*
South San Francisco, CA · Arrayed Screening, Neuroscience

- Adapted high-content phenotypic assays for high-throughput small-molecule screening (tuberous sclerosis complex, ALS) on near-fully-automated platforms (BioTek, High-Res).
- Built a digital phase-contrast timecourse imaging modality and Python analysis suite to track neuron growth dynamics and drive label-free phenotypic discovery.

Technical Associate II, Massachusetts Institute of Technology *2021 – 2022*
Cambridge, MA · Synthetic Neurobiology Group (Boyden Lab)

- Optimized multiplexed expansion revealing (mExR) stripping for 7+ rounds of IHC on 16× linearly expanded specimens, enabling nanoscale visualization of 30+ proteins (Nature Communications, 2024).
- Developed and executed image-analysis pipelines in ImageJ/Fiji and MATLAB.

Associate Scientist II, AbbVie *2018 – 2021*
Cambridge, MA · Foundational Neuroscience Center, Early R&D

- Built primary murine and hiPSC neuronal tau-aggregation high-content screening assays (Revvity Opera, Harmony).
- Modeled intercellular tau propagation in Alzheimer's disease using microfluidic devices; authored R pipelines for reproducible dose-response profiling (Journal of Biological Chemistry, 2025).

Research Technician I, Massachusetts General Hospital *2017 – 2018*
Boston, MA

- Used voltage-clamp electrophysiology in GABA-receptor-expressing *Xenopus* oocytes to elucidate mechanisms of general anesthetics (European Journal of Medicinal Chemistry, 2019).

EDUCATION

Harvard University — Graduate coursework, Extension School *2017 – 2020*
Completed while employed full-time: Social Medicine; Introduction to Programming (Python); Image Processing & Computer Vision (Python).

B.A., Colgate University, *cum laude* *2013 – 2017*
Major: Cellular Neuroscience. Minor: Mathematical Systems Biology.

PEER-REVIEWED PUBLICATIONS

1. Yanamandra K, Romanul N, Patel H, Sanders H, Chang R, Darcy A, Brown N, Kane C, Wu HY, Jagtap K, Chancellor S, Morozova O, Xi M, Hogan C, **Titterton K**, Wu JW, Manos J, Karran E, Dellovade T, Langlois X. Conformation-specific anti-pathological tau antibody raised against AD brain-derived seed-competent tau reduces tau pathology in transgenic mice. *Mol Neurodegener Adv*. 2026;2(1):43. doi:10.1186/s44477-026-00049-6.
2. Sebastian RM, Kupp R, Nanavati D, Parfentev I, Krzystek TJ, Romanul N, Yanamandra K, Preiss CN, Manos JD, Wu HY, Volkova A, Crescenti S, **Titterton K**, Welker AM, Townsend M, Karran E, Langlois X, Wu JW. SynCAR platform for capturing neuronal synaptic proteopathic seeds. *iScience*. 2026;29(7):116362. doi:10.1016/j.isci.2026.116362. PMID 42325569; PMCID PMC13276770.
3. Ewald JD, **Titterton KL**, Bäuerle A, Beatson A, Boiko DA, Cabrera ÁA, Cheah J, Cimini BA, Gorissen BL, Harrill J, Jones TR, Karczewski KJ, Crute CE, Rouquie D, Seal S, Weisbart E, White B, Carpenter AE, Singh S. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Syst*. 2026;17(5):101566. doi:10.1016/j.cels.2026.101566. PMID 41903531.
4. Seal S, Dee W, Shah A, Cerisier N, Zhang A, Miglietta E, **Titterton K**, Cabrera ÁA, Boiko D, Beatson A, Slabaugh G, Taboureau O, Carreras Puigvert J, Singh S, Spjuth O, Bender A, Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*. 2026;17(1). doi:10.1038/s41467-026-68725-5. PMID 41651839; PMCID PMC12988037.
5. **Wu JW, Titterton K**, Sebastian R, Romanul N, Yanamandra K, Mastis B, Chang R, Khasnavis S, Kwon T, Chai Y, Manos JD, Preiss CN, Hu M, Welker AM, Liao F, Dellovade T, Karran E, Langlois X. Development of a robust, physiologically relevant neuronal tau aggregation assay for tau-related drug discovery. *J Biol Chem*. 2025;301(12):110853. doi:10.1016/j.jbc.2025.110853. PMID 41135680; PMCID PMC12670552.
6. Kang J, Schroeder ME, Lee Y, Kapoor C, Yu E, Tarr TB, **Titterton K**, Zeng M, Park D, Niederst E, Wei D, Feng G, Boyden ES. Multiplexed expansion revealing for imaging multiprotein nanostructures in healthy and diseased brain. *Nat Commun*. 2024;15(1):9722. doi:10.1038/s41467-024-53729-w. PMID 39521775; PMCID PMC11550395.

7. Manos JD, Preiss CN, Venkat N, Tamm J, Reinhardt P, Kwon T, Wu J, Winter AD, Jahn TR, Yanamandra K, **Titterton K**, Karran E, Langlois X. Uncovering specificity of endogenous TAU aggregation in a human iPSC-neuron TAU seeding model. *iScience*. 2022;25(1):103658. doi:10.1016/j.isci.2021.103658. PMID 35072001; PMCID PMC8761709.
8. Wu B, Jayakar SS, Zhou X, **Titterton K**, Chiara DC, Szabo AL, Savechenkov PY, Kent DE, Cohen JB, Forman SA, Miller KW, Bruzik KS. Inhibitable photolabeling by neurosteroid diazirine analog in the β3-subunit of human heteropentameric type A GABA receptors. *Eur J Med Chem*. 2019;162:810-824. doi:10.1016/j.ejmech.2018.11.019. PMID 30544077; PMCID PMC6912599.

CONFERENCE PRESENTATIONS & POSTERS

1. **Titterton K**, Boiko D, Cabrera ÁA, Bäuerle A, Tyagi A, Saneinejad S, Beatson A, White B. "Accurate and interpretable in silico clinical risk assessment for drug-induced liver injury (DILI) from molecular structure." *Society of Toxicology (SOT) Annual Meeting*, 2025.
2. **Titterton K**, et al. "Development of a tau propagation neuronal microfluidic cell culture model for assessing inhibitors of tau propagation." *Society for Neuroscience (SfN) Global Connectome Virtual Meeting*, January 2021.
3. **Titterton K**, et al. "High content analysis (HCA) of tau aggregation in primary neurons: screening anti-tau antibodies and modeling tauopathy progression in vitro." *SBI2 High Content Meeting*, September 2019, Harvard Medical School, Boston, MA.

Earlier research presentations

- "Investigating RNAi as a potential therapeutic for Huntington's Disease in a Transgenic mHTT145Q Zebra Finch (*Taeniopygia guttata*) Model." Colgate University Neuroscience Thesis Poster Presentations, 2017.
- "Atopy and helminth infection in Ethiopian schoolchildren." Colgate Summer Research Poster Symposium, 2016.
- "PDE2 regulation of β-arrestin signaling." Systems Biology Center New York (SBCNY) Summer Trainee Poster Symposium, 2015, Icahn School of Medicine at Mount Sinai.

HONORS & AWARDS

- 2nd Place, **BioHack NYC 2023** — OpenTronsformers (LLM-driven lab-robot automation).
- *cum laude*, Colgate University, 2017.
- Dorothy-Waltham Award for Athletic Leadership, Colgate University, 2017.

PROFESSIONAL MEMBERSHIPS

- Society of Toxicology (SOT); Society for Neuroscience (SfN); Society of Biomolecular Imaging and Informatics (SBI2).

SELECTED PROJECTS & CREATIVE WORK

Bare Metal — first solo IRL art exhibit, Human Tech Week, San Francisco *2025*

Vivarium 2.0 — automated poison-dart-frog habitat with misting + motion-triggered video capture *2024*

OpenTronsformers — LLM code-generation companion for OT-2 lab robots (2nd place, BioHack NYC) *2023*

TECHNICAL SKILLS

Computational: LLM codegen (Claude Code, Cursor); image embeddings (visualization, clustering, feature derivation, normalization); Python (data science, image processing & segmentation; Jupyter, NumPy, SciPy, scikit-learn, pandas, matplotlib, seaborn, xlsxwriter, Colab); R (ggplot2, tidyverse, drc, platetools); dose-response curve fitting (GraphPad Prism, R, Python); univariate & multivariate statistics; design of experiments (DOE); ImageJ/Fiji macro scripting; Benchling ELN.

Wet lab: Cell culture (primary murine neuron, hiPSC neuron models); microfluidic cell-culture models; ICC / IHC; confocal microscopy (manual, PE Operetta / Opera Phenix); high-content image analysis (PerkinElmer Harmony); automated liquid handling & schedulers (OpenTrons, Integra Assist Plus / Viaflo, BioTek, Labcyte Echo, High-Res Biosolutions, Green Button Go); assay development & HT/HC screening; stereotaxic surgery / tissue microdissection; voltage-clamp electrophysiology.

CERTIFICATIONS

- FII Level 1 Freediving (October 2019); Molchanov Level 2 Freediving (October 2025); SII Shark Ecology Specialty (April 2020).